

Algorithms

Lecture #10

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Algorithm

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Medians and Selection

Medians and Selection

- The selection problem can be easily solved by simply sorting the numbers of A and returning $A[k]$.
- Sorting, however, requires $\Theta(n \log n)$ time.
- The question is: can we do better than that?
- In particular, is it possible to solve the selections problem in $\Theta(n)$ time?
- The answer is **yes**. However, the solution is far from obvious.

Medians and Selection

Medians and Selection

- We will use a variation of the divide and conquer strategy called the *sieve technique*.
- In divide and conquer, we break the problem into a small number of smaller subproblems which we solve recursively.

Medians and Selection

Medians and Selection

- In the selection problem, we are looking for an item.
- We will divide the problem into subproblems.
- However, we will **discard** those small subproblems for which we determine that they do not contain the desired answer (*the variation*).

Medians and Selection

Medians and Selection

Here is how the sieve technique will be applied to the selection problem.

- We will begin with the given array $A[1..n]$.
- We will pick an item from the array, called the *pivot element* which we will denote by x .
- We will talk about how an item is chosen as the pivot later; for now just think of it as a random element of A .

Medians and Selection

Medians and Selection

We partition A into three parts.

1. $A[q]$ contains the pivot element x ,
 2. $A[1..q - 1]$ will contain all the elements that are less than x and
 3. $A[q + 1..n]$ will contains all elements that are greater than x .
- Within each sub array, the items may appear in any order.

Partitioning

Partitioning

$A[p..r]$ partitioned about the pivot x .

				pivot			
				↓			
p							r
5	9	2	6	4	1	3	7
p			q				r
3	1	2	4	6	9	5	7
< x		x		> x		x	

Before partitioning

After partitioning

Medians and Selection

Medians and Selection

- The rank of the pivot x is $q - p + 1$ in $A[p..r]$.
- Let $\text{rank}_x = q - p + 1$.
- If $k = \text{rank}_x$ then the pivot is k^{th} smallest.
- If $k < \text{rank}_x$ then search $A[p..q - 1]$ recursively.

Medians and Selection

Medians and Selection

- If $k = \text{rank}_x$ then the pivot is k^{th} smallest.
- If $k < \text{rank}_x$ then search $A[p..q - 1]$ recursively.
- If $k > \text{rank}_x$ then search $A[q + 1..r]$ recursively. Find element of rank $(k - q)$ because we eliminated q smaller elements in A .

Select Algorithm

Select Algorithm

```
SELECT( array A, int p, int r, int k)
1 if (p = r)
2   then return A[p]
3 else x ← CHOOSE PIVOT(A, p, r)
4   q ← PARTITION(A, p, r, x)
5   rank_x ← q - p + 1
6   if k = rank_x
7     then return x
8   if k < rank_x
```

Select Algorithm

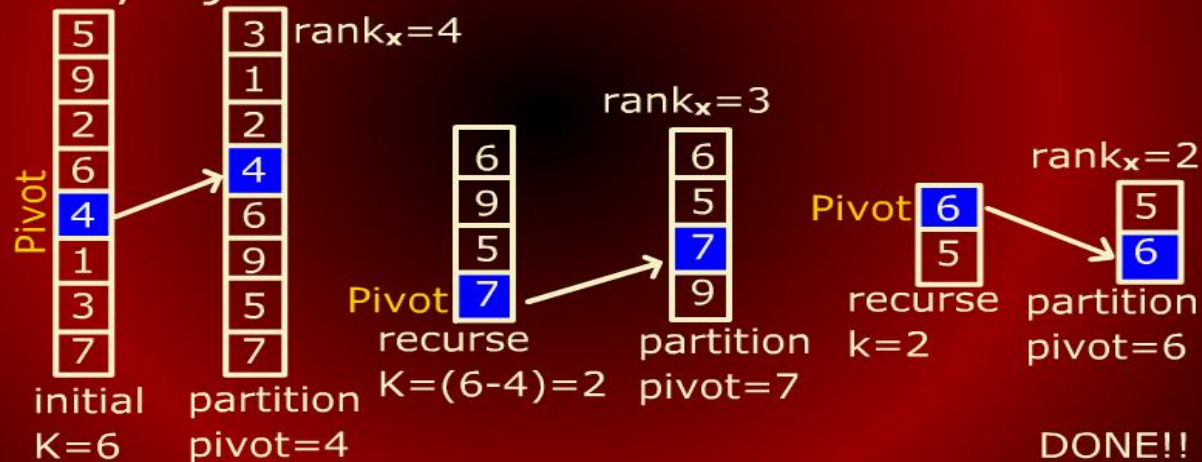
Select Algorithm

```
5  rank_x  $\leftarrow$  q - p + 1
6  if k = rank_x
7    then return x
8  if k < rank_x
9    then return SELECT(A, p, q-1, k)
10 else return SELECT(A, q+1, r, k-q)
```

Select Algorithm

Select Algorithm

- Example: select the 6th smallest element of the set {5, 9, 2, 6, 4, 1, 3, 7}



Medians and Selection

Medians and Selection

- We will discuss how to choose a pivot and the partitioning later.
- For the moment, we will assume that they both take $\Theta(n)$ time.
- How many elements do we eliminate in each time?

Steve Technique

Design and Analysis of Algorithm